

E-Cigarette Supply: A Conference-Version Model

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Draft

*Motivating facts, model environment, and equilibrium definition for the internal-conference draft.
Static pricing, exogenous state, explicit fringe.*

1 Motivating facts

The tracked e-cigarette market is small- N and its leadership turns over completely. Figure 1 plots annual, mL-weighted brand composition from the Nielsen RMS panel: the top-5 brands each year, named and individually shaded, with the market leader starred, and everyone else pooled into “Other.”

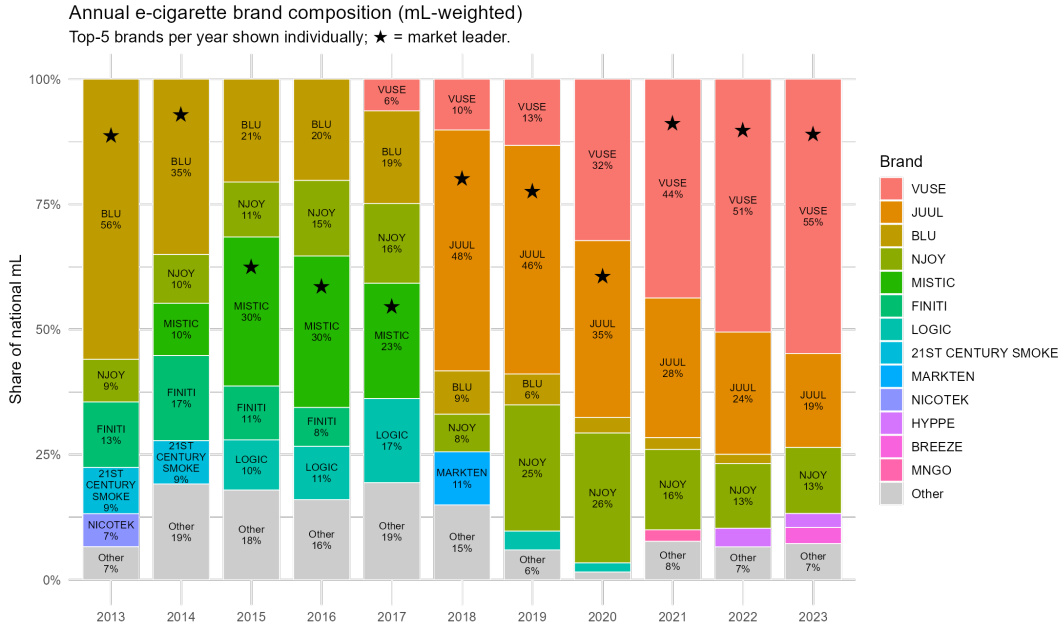


Figure 1: Annual e-cigarette brand composition, mL-weighted, 2013–2023. Source: Nielsen RMS retail scanner panel (national, annual, mL-weighted). Top-5 brands per year shown individually; ★ = market leader (CR_1). Rest = “Other.”

Three things in this figure motivate the model form directly:

1. **A small number of firms account for most of the market, every year.** CR_3 never drops below 55.5% (2015) and reaches 87.9% (2022) — a handful of firms, not a competitive fringe of many, set the terms of competition in any given period. This motivates capping the dynamic game at $\bar{N}$ active firms rather than modeling every brand as a strategic player.

2. **The identity of the leader changes completely, more than once.** BLU (2013) → MISTIC (2015–2017) → JUUL (2018–2020) → VUSE (2021–2023) — three full leadership transitions in an eleven-year panel. A static entry-barriers story doesn’t produce this; a dynamic model with state-dependent entry, exit, and (eventually) investment does.
3. **The residual below CR_3 is not a fixed constant.** It ranges from 12–22% of national mL in most years down to 12.3% in 2020 — small but persistent, and volatile enough that it needs to be modeled explicitly rather than assumed away (Table 1; see also the firm-count/materiality diagnostics in the companion 08.27.2026 note, which show the sub- CR_3 group holding 15–22% of national mL through 2014–2018 specifically).

Table 1: Annual concentration, mL-weighted national brand shares.

Year	Leader	Leader share	Runner-up	CR_3	HHI
2013	BLU	56.0%	FINITI	78.3%	3519
2014	BLU	35.0%	FINITI	61.8%	1869
2015	MISTIC	24.0%	BLU	55.5%	1430
2016	MISTIC	25.9%	BLU	61.3%	1564
2017	MISTIC	20.8%	BLU	56.0%	1430
2018	JUUL	46.9%	VUSE	67.7%	2612
2019	JUUL	42.3%	NJOY	77.0%	2609
2020	JUUL	33.7%	VUSE	87.3%	2639
2021	VUSE	43.7%	JUUL	87.7%	2966
2022	VUSE	50.5%	JUUL	87.9%	3338
2023	VUSE	54.8%	JUUL	86.8%	3553

HHI is U-shaped over the panel — highly concentrated in 2013 ($\sim 3,500$), falls to the DOJ’s moderately-concentrated floor by 2015–2017 ($\sim 1,400$ – $1,600$) as cigalike incumbents fragment, then re-concentrates from 2018 on as first JUUL and then VUSE consolidate the market. That *U*-shape is itself evidence against a static model: concentration is an equilibrium outcome of entry, exit, and (later) investment responding to shocks — first the salt-nicotine innovation, then a wave of FDA actions in 2019–2020 — not a fixed structural feature of the industry.

2 Environment

Players and timing, period t (monthly). Up to $\bar{N} = 5$ incumbent firms are active, indexed $i = 1, \dots, n_t \leq \bar{N}$, plus one explicit competitive fringe (a single aggregated player, not $\bar{N}$ -capped) and at most one potential entrant. Each period: (i) states are realized; (ii) active firms and the fringe compete in a static pricing subgame; (iii) profits realize; (iv) incumbents decide continue/exit and the potential entrant decides enter/stay out; (v) states evolve into $t + 1$.

State. Firm i carries $\omega_{i,t} = (a_{i,t}, \xi_{i,t})$ — an accumulated-quality index and an idiosyncratic residual shock, both firm-and-time-indexed. In this version both are **exogenous**, evolving as first-order Markov processes estimated directly off the data:

$$a_{i,t+1} \sim F_a(\cdot \mid a_{i,t}), \quad \xi_{i,t+1} \sim F_\xi(\cdot \mid \xi_{i,t}).$$

The fringe carries a single aggregate state $a_{F,t}$, exogenous, estimated the same way off the fringe’s observed aggregate series (net of the 2021 Nielsen-panel-onboarding correction — see the companion model note, §4.6).

3 Demand

A firm-level nested logit: every active firm’s product and the fringe sit in one “e-cigarette” nest against a single outside-option nest. Firm i ’s mean utility is

$$\delta_{i,t} = \beta_a a_{i,t} + \xi_{i,t} - \alpha p_{i,t},$$

where β_a is a free coefficient on the quality state (not normalized to 1 — $a_{i,t}$ is an observable, physically-scaled quantity, and the Ericson–Pakes (1994) canonical single-period profit function gives its own quality term a free coefficient rather than unit weight) and $\xi_{i,t}$ is the exogenous residual already carried in the state $\omega_{i,t} = (a_{i,t}, \xi_{i,t})$, included per standard mean-utility convention (Berry 1994/BLP). Market shares follow the standard nested-logit form with nest parameter $\sigma \in [0, 1)$ governing within-nest substitution. (Nest correlation σ , the price coefficient α , and β_a are estimated objects, not calibrated here.)

Price construction. For each eligible UPC u in month t , compute $p_{u,t} = \text{real revenue}_{u,t} / \text{mL sold}_{u,t}$ after collapsing any T2 rows. Firm price $p_{i,t}$ is the simple, unweighted mean across its UPCs, using the same sample as $a_{i,t}$ (positive sales, at least three stores, UNKNOWN excluded). This may matter later for profit measurement: $p_{i,t}$ times firm mL need not equal observed revenue.

Instruments. Two regressors in the linear estimating equation (companion solution note, §1) are endogenous — $p_{i,t}$ and the within-nest-share term $\sigma \ln(s_{i,t}/s_{g,t})$ — and are instrumented. For the within-nest-share term, the instrument is the number of firms active in the market that period: more active rivals mechanically compress a firm’s within-nest share without being correlated with that firm’s own $\xi_{i,t}$. For price, the instrument is a population-weighted national excise-tax rate built from state-month e-liquid excise taxes (\$/mL): $\text{TaxIV}_t = \sum_s \frac{\text{Pop}_s}{\text{Pop}_{US}} \cdot \text{tax}_{s,t}$, aggregating up to a single national series rather than a firm-specific shift-share design (per Marc’s recommendation, in place of an earlier plan to weight state taxes by each firm’s own baseline state-sales shares).

4 Static pricing subgame

Given the period’s active-firm set and states, each firm sets price to maximize static profit:

$$\pi_i(\omega_t) = \max_{p_i} (p_i - c(\omega_{i,t})) q_i(p_t, \omega_t) - FC, \quad q_i(p_t, \omega_t) = M_t \cdot s_i(p_t, \omega_t),$$

where q_i is quantity and s_i is the nested-logit market share from §3; market size M_t cancels out of the resulting first-order condition, so the equilibrium price depends only on shares. The Bertrand–Nash equilibrium price vector solves the resulting system of first-order conditions simultaneously across all active firms (and the fringe, if it prices rather than holding a reduced-form price) — a standard logit markup condition, expanded and solved numerically in the companion solution note.

This is genuinely static: p_t^* depends only on the current-period state ω_t , not on any continuation value. “Price” varies dynamically across the paper’s periods only because the state ω_t on which it is conditioned is itself evolving — not because pricing today is chosen with tomorrow in mind.

5 Dynamic entry and exit

Let $\pi_i^*(\omega_t)$ denote the equilibrium static profit from §4. An incumbent’s value function:

$$V_i(\omega_{i,t}, \omega_{-i,t}) = \pi_i^*(\omega_t) + \max \left\{ \underbrace{\phi_i}_{\text{exit: scrap value}}, \underbrace{\beta \cdot \mathbb{E}[V_i(\omega_{i,t+1}, \omega_{-i,t+1}) \mid \omega_t, \text{continue}]}_{\text{continue}} \right\}.$$

Optimal exit rule: firm i exits iff $\phi_i > \beta \cdot \mathbb{E}[V_i(\omega_{i,t+1}, \omega_{-i,t+1}) \mid \omega_t, \text{continue}]$. Define the implied exit indicator:

$$\lambda^{\text{exit}}(\omega_{i,t}; \omega_{-i,t}) = \mathbf{1}\{\phi_i > \beta \cdot \mathbb{E}[V_i(\omega_{i,t+1}, \omega_{-i,t+1}) \mid \omega_t, \text{continue}]\}.$$

The potential entrant compares its expected discounted value of entering against a sunk entry cost x_e ; it enters iff the former net of the latter is non-negative. Define the implied entry indicator:

$$\lambda^{\text{entry}}(\omega_t) = \mathbf{1}\{\beta \cdot \mathbb{E}[V_e(\omega_{e,0}, \omega_{-e,t+1}) \mid \omega_t] - x_e \geq 0\}.$$

(λ , not η — in Ericson–Pakes’ own model η' already denotes something unrelated, an aggregate industry-wide shock inside the transition kernel, and χ is claimed too, as the exit-decision indicator itself in the closely-related Doraszelski–Pakes–Satterthwaite restatement of this model. λ sidesteps both of those as well as the nested-logit correlation parameter σ already used in §3.) Equilibrium is a fixed point in $(\lambda^{\text{exit}}, \lambda^{\text{entry}})$ and the value function, computed via the standard Pakes–McGuire (1994) iteration — full algorithm, plus the transition-probability mechanics these indicators feed into, in the companion solution note.

Does an incumbent factor in a rival’s possible exit, or a new entrant? Yes, and this is already fully active in the present version, even before investment becomes endogenous. The value function $V_i(\omega_{i,t}, \omega_{-i,t})$ explicitly takes rivals’ states $\omega_{-i,t}$ as an argument, and the continuation value $\mathbb{E}[V_i(\omega_{i,t+1}, \omega_{-i,t+1}) \mid \omega_t, \text{continue}]$ integrates over the full future state vector, which includes what rivals do. Since ϕ_j and x_e are private draws, firm i cannot know for certain whether rival j will exit or whether the entrant will show up — but it knows the equilibrium cutoff rules (λ^{exit} exit iff ϕ_j exceeds some threshold given j ’s state; λ^{entry} enter iff expected value exceeds x_e), so it can compute the probability of each, and its own continuation value is an expectation over all of that. “Equilibrium is a fixed point in $(\lambda^{\text{exit}}, \lambda^{\text{entry}})$ and the value function” above is exactly this mutual consistency — every firm’s best response already correctly anticipates every other firm’s probabilistic policy.

Exit sample note. MarkTen’s and MISTIC’s exits track dated parent-company decisions (Altria discontinuing MarkTen in 2018; Ballantyne winding MISTIC down ~2020) rather than the state-dependent comparison above, and are carved out as exogenous ownership shocks; the remaining top-5 exits are the identifying variation for ϕ_i .

6 The fringe

The fringe enters the demand system as one more alternative inside the nest, $\delta_{F,t} = \beta_a a_{F,t}$ (the same demand coefficient β_a as the $\bar{N} = 5$ firms; the fringe carries one aggregate state, not a firm-and-time-indexed pair, so there is no separate $\xi_{F,t}$), competing for share against the $\bar{N} = 5$ firms and the outside option in the same static pricing solve. It carries no Bellman equation in this version — no entry/exit decision of its own, just a correctly time-varying aggregate presence. This is the cheap version, chosen specifically to avoid adding dimensions back into the dynamic game’s state space right after the state-space size was the reason to move to Experience-Based Equilibrium in the first place (§8).

7 What this version leaves for later

This is a conference draft, not the paper’s final specification. Explicitly deferred:

- **Investment.** $a_{i,t}$ is exogenous here. The full model endogenizes it via a binary invest/don’t-invest choice ($d_{i,t} \in \{0, 1\}$, a firm-and-time-varying investment cost $\kappa_{\text{inv},i,t}$ if investing, stochastic success), solved as an Experience-Based Equilibrium rather than a full Markov Perfect Equilibrium because the state space (up to 5 firms $\times$ 2 states each) is too large to solve directly — per Ariel Pakes’s recommendation. $\kappa_{\text{inv},i,t}$ is also where FDA regulation

enters the full model (stepping up at known enforcement dates, with an additional JUUL-specific increment) — chosen over directly capping $a_{i,t}$ because it lets rivals' continuation values respond to a regulation-weakened competitor, which a static/capped-state version cannot do.

- **The cost channel for $a_{i,t}$.** Dropped. Firm-level learning-curve regressions find no price-identified cost-reduction signature for the pod-era leaders (JUUL/VUSE/NJOY) — only for the 2013–2017 cigalike cohort — and price alone can't separate cost movements from markup movements without additional data (margins, compliance costs) this draft doesn't have. $a_{i,t}$ is carried here purely as a demand-side (mean-utility) state.
- **Question : does crossing the incumbent-share threshold count as entry/exit?** Demand-side incumbent status is now defined by annual market share rather than the $\bar{N} = 5$ cap directly: a firm is treated as a full incumbent in the nested logit for a given calendar year iff its annual market share exceeds 7% (the logit itself stays at monthly frequency; only the incumbent/fringe assignment is fixed per year), with firms below that line folded into the fringe aggregate for that year. This is now decoupled from $\bar{N} = 5$, which governs only the dynamic entry/exit game. Open question: when a firm crosses the threshold — in either direction — does that reclassification itself count as dynamic entry or exit (i.e., does it trigger the $\lambda^{\text{exit}}/\lambda^{\text{entry}}$ comparison against ϕ_i/x_e), or is entry/exit reserved for a firm genuinely starting or stopping sales altogether, independent of this share-threshold reclassification? The two interpretations imply different identification strategies for ϕ_i and x_e . At 7%, this is moot in the current data: every firm has a single unbroken incumbent spell (NJOY included, continuous 2013–2023), and $\bar{N} = 5$ never binds either — the annual incumbent count ranges 3–6, hitting 6 in 2014 and again in 2018 as the cigalike-era leaders and the pod-era entrants briefly overlap above the line. **Note: NJOY becomes an issue at a higher threshold.** At 10% or above, NJOY drops below the cutoff in 2018 (7.6% share) and returns from 2019 on — a genuine gap-and-return case, so the entry/exit interpretation question stops being moot and gets entangled with the choice of cutoff itself.
- **A strategic fringe.** The fringe's aggregate state is exogenous here. A richer version replaces it with a monopolistically-competitive free-entry sector whose size responds endogenously to the same shocks hitting the top firms.
- **Question : legacy-brand bias in $a_{F,t}$.** $a_{F,t}$ is an unweighted average of nicotine content across active fringe UPCs each month. Diagnostics find that once a tracked firm permanently exits the top-5 roster, its residual UPCs can systematically bias $a_{F,t}$ if that firm's own nicotine level is a persistent outlier relative to the rest of the fringe — confirmed for two firms (MarkTen, pulling $a_{F,t}$ up ~ 12 – 20% through 2019; NICOTEK, pulling it down ~ 10 – 11%), with every other tracked firm's post-exit residual found negligible. A general correction is derived and validated — exclude a firm's post-exit UPCs from the fringe average once its deviation from that month's fringe mean exceeds $\sim 10\%$, applied only to months after a firm's last incumbent year so it never touches a firm's pre-incumbency rise (this distinction matters: a naive version of the rule, applied without the pre/post split, would have wrongly flagged VUSE's and JUUL's own early, genuinely-rising histories) — but not yet implemented. Open question: is this worth correcting in the construction, or is the resulting bias small enough relative to everything else in the model to leave as a documented limitation?

- **What “cost of nicotine” is a cost of, functionally,** and the ownership-vs-quality question as a driver of turnover (§7.6 of the companion diagnostics note) — both flagged, neither resolved, neither needed for this draft.

8 Framework and references

Ericson & Pakes (1995) for the entry/exit/investment framework; Pakes & McGuire (1994, 2001) for the computational algorithm (the second paper’s recurrent-states restriction is the direct precursor to the EBE move); Spence (1981) for the learning-by-doing idea originally considered for $a_{i,t}$ ’s state interpretation (dropped — see §7), with Benkard (2000, 2004) as the specific aircraft-manufacturing application of that idea, not its origin; Fershtman & Pakes (2012) for Experience-Based Equilibrium. Citations characterized from memory at the level of what each paper does — worth confirming exact notation against the originals before this goes into a submitted draft.